

Sarvagya Jaiswal

+91-7081282726 | jaiswal.sarvagya1@gmail.com | linkedin.com/in/sarvagyajaiswal | github.com/Sarvjais12 | sarvjais12.github.io | huggingface.co/Sarvjais12

EDUCATION

KIIT University

Bachelor of Technology in Computer Science, CGPA: 8.14/10.0

Bhubaneswar, India

2023 – 2027 (Expected)

EXPERIENCE

Infiltrix, Early-Stage AI Cybersecurity Startup

Remote

Machine Learning Engineer, Founding Team

Dec 2025 – Present

- Architected the Fast Anomaly Gate, a hybrid detection engine fusing Random Forest, Autoencoder, and Logistic Regression with trained OR-gate fusion; processed 2.8 million CICIDS2017 network flows (77 features, 15 attack types), achieving **99% accuracy, 94% attack recall, and 0.1% false containment rate** at 50–76ms inference latency.
- Containerized the pipeline as a FastAPI service using Docker, Redis, and PostgreSQL; built Z-score statistical drift detection, benign-calibrated threshold selection, and a Containment Health Score metric; identified and resolved temporal data leakage and autoencoder normalization bugs in production.

Fox Trading, Wipro Affiliated

Remote

Machine Learning Intern

May 2025 – Jun 2025

- Built an ensemble Diabetes Prediction model using Random Forest, XGBoost, and SHAP-based feature importance, reducing preprocessing-to-deployment latency from 48 hours to under 4 hours; benchmarked Random Forest, Support Vector Machine, and Logistic Regression on mHealth sensor data to classify 12 distinct physical activities.
- Implemented a Bike Sharing Demand forecasting pipeline using temporal and weather features via regression analysis, optimizing resource allocation by 25%.

PROJECTS

Skyclad RAG Agent | LangGraph, ChromaDB, BM25, Groq | GitHub | Demo

2025

- Built an agentic Retrieval-Augmented Generation system over 100 arXiv papers with **5-way large language model routing** (retrieve, tool, clarify, refuse, answer) via LangGraph conditional edges; integrated a live arXiv API tool for out-of-corpus queries and a rolling semantic memory summarizer for context-aware query rewriting.
- Implemented hybrid sparse-dense retrieval fusing BM25 and vector search via Reciprocal Rank Fusion; ablation across 6 queries showed hybrid consistently outperforms sparse-only on paraphrase queries. Scored **36 out of 36 (100%)** on a hand-written routing evaluation harness covering all 5 action paths.

Agentic Multi-Modal Assistant | FastAPI, Anthropic API, Tesseract OCR, Whisper | GitHub

2025

- Designed a fully asynchronous FastAPI backend to handle Text, Image OCR, PDF, and Audio inputs without blocking the event loop; built an intent-routing orchestrator with a clarification gate, per-token cost estimator, and PDF-to-OCR fallback.
- Shipped with modular agent and tasks architecture, a system architecture diagram, and **42 pytest test cases** covering all input modalities and edge cases.

Spatial-Temporal Deepfake Detector | PyTorch, EfficientNet-B0, Bi-LSTM, OpenCV | GitHub | Demo

2025

- Built a hybrid CNN-RNN architecture where EfficientNet-B0 extracts 1,280-dimensional spatial features per frame and Bi-LSTM models temporal inconsistencies, catching micro-expression jitter invisible to frame-by-frame classifiers. Achieved **94% accuracy** on adversarial GAN manipulation; attention mechanism highlights manipulated frames. Live on Hugging Face Spaces.

ContentFlow, Social Media Content Planner | Flutter, Dart, Firebase, Firestore, Provider | GitHub

2025

- Full-stack cross-platform mobile application with Firebase Authentication, real-time Firestore database, multi-platform post scheduling (Instagram, LinkedIn, Twitter) with character limit enforcement per platform, hashtag manager, and date/time scheduler across 8 screens using MVVM architecture.

TECHNICAL SKILLS

Languages: Python, Dart, C, C++, Java

Machine Learning and AI: PyTorch, TensorFlow, Scikit-Learn, XGBoost, LangChain, LangGraph, FAISS, ChromaDB, Hugging Face Transformers, OpenCV, Pandas, NumPy

Backend and Mobile: FastAPI, Flutter, Firebase, Docker, Redis, PostgreSQL, GitHub Actions, Gradio, Pytest

Domains: Computer Vision, Natural Language Processing, Retrieval-Augmented Generation, Anomaly Detection, Mobile Development, Async Backend Systems

TECHNICAL WRITING

4 published articles on machine learning architecture and MLOps (dev.to and Medium): deepfake detection, LangGraph RAG agents, Grad-CAM medical AI, and Docker containerization. Research paper on AQI anomaly detection using Isolation Forest on multivariate time-series data, in preparation.